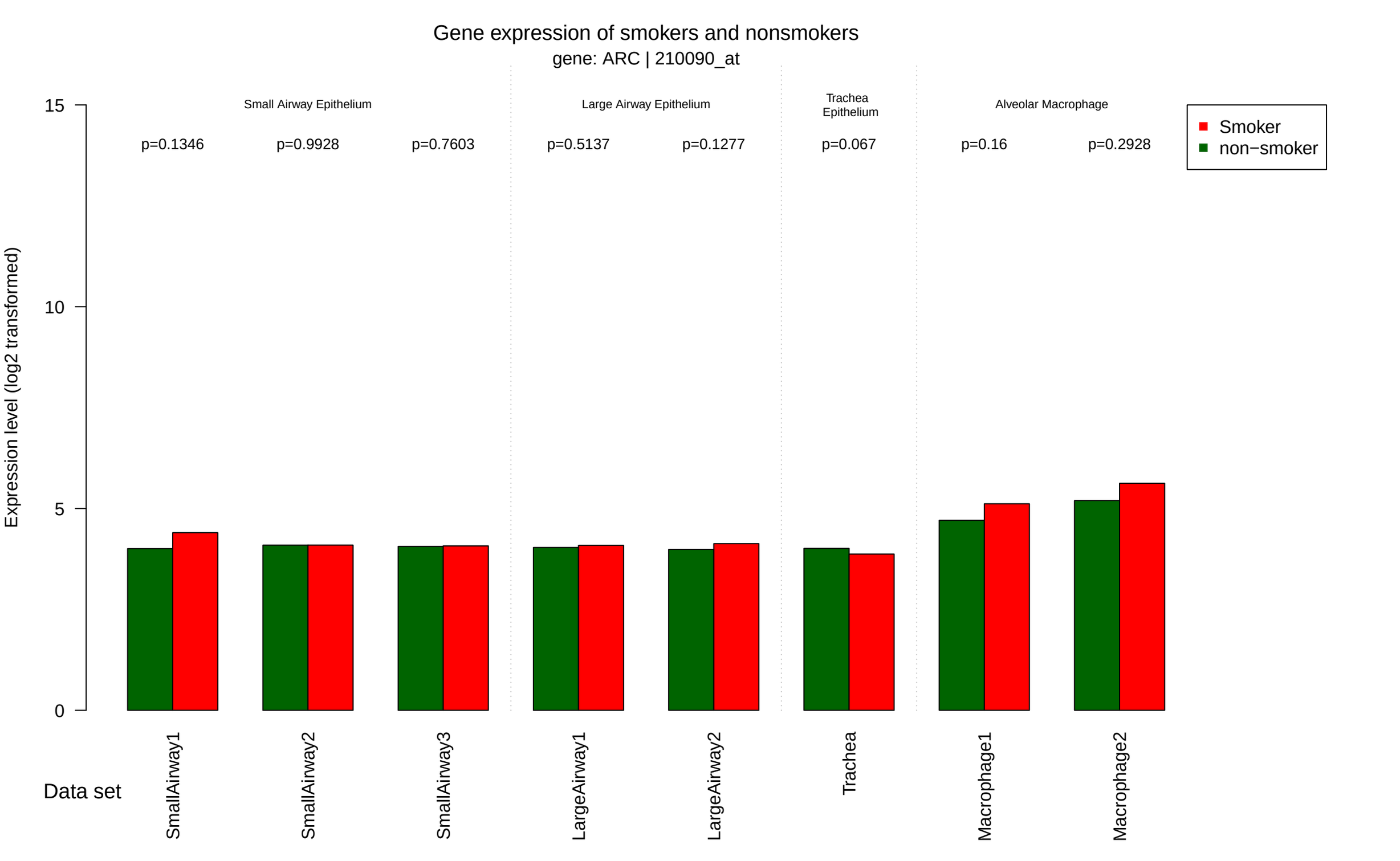
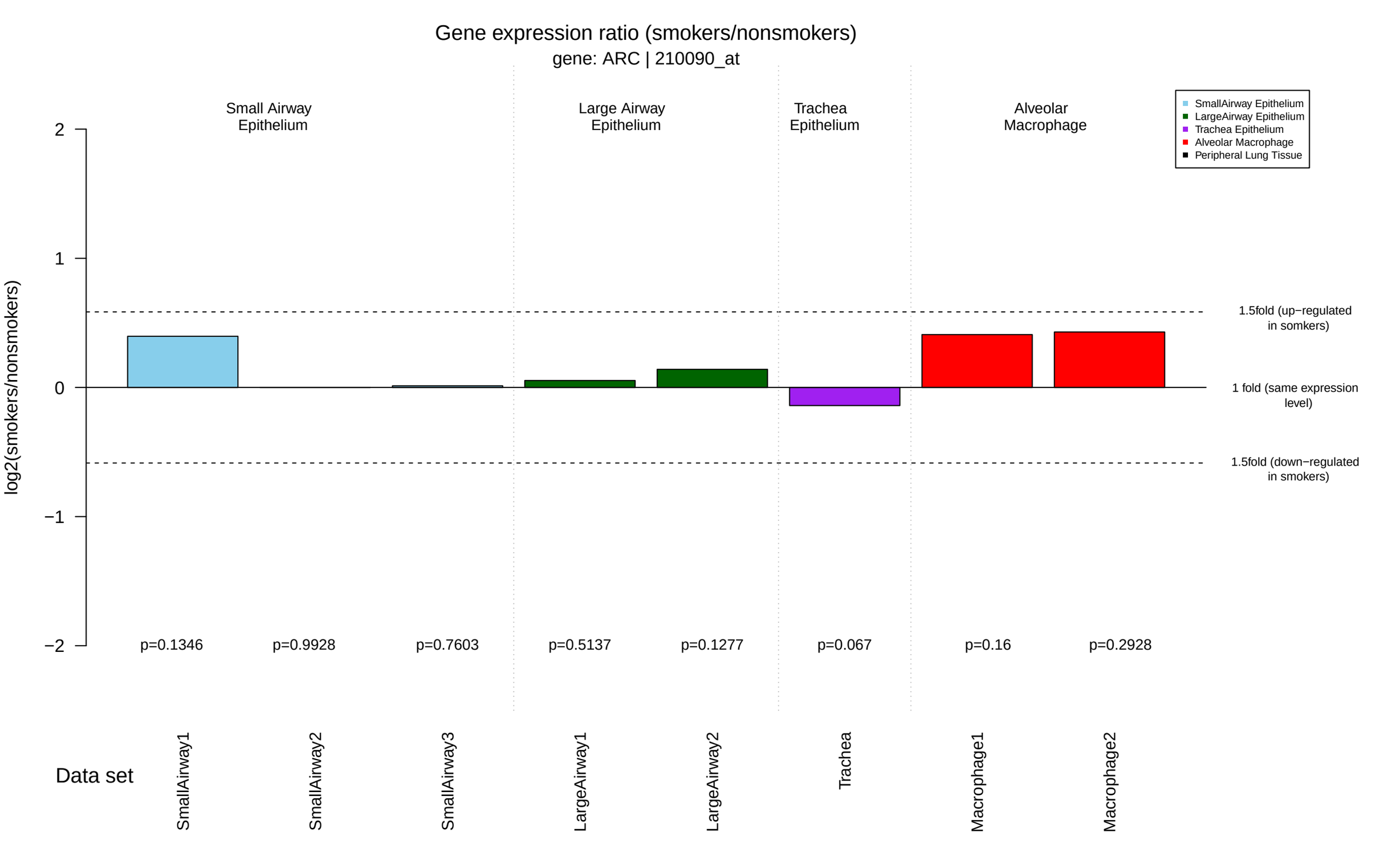
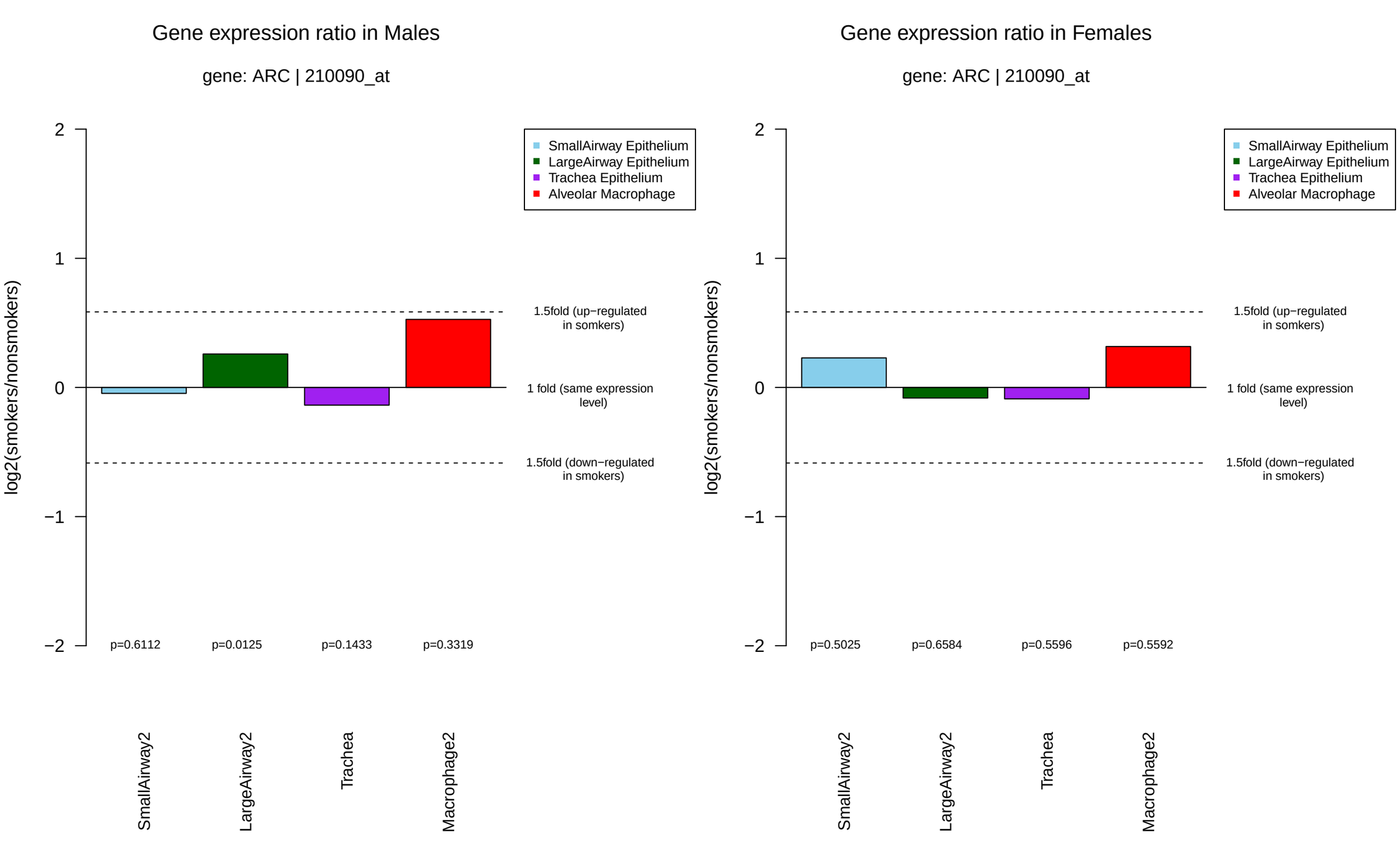
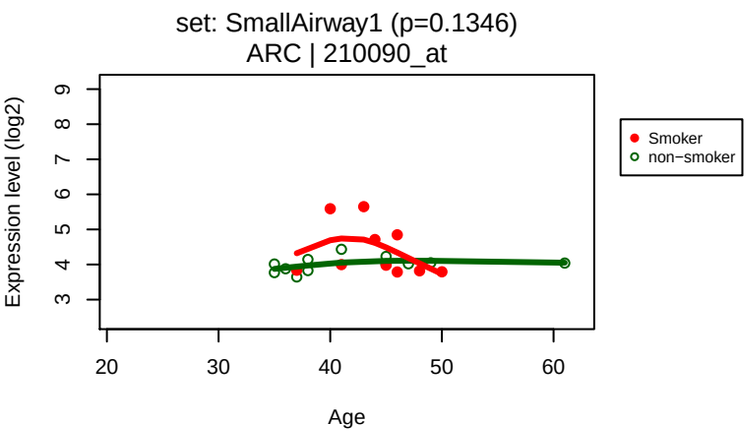
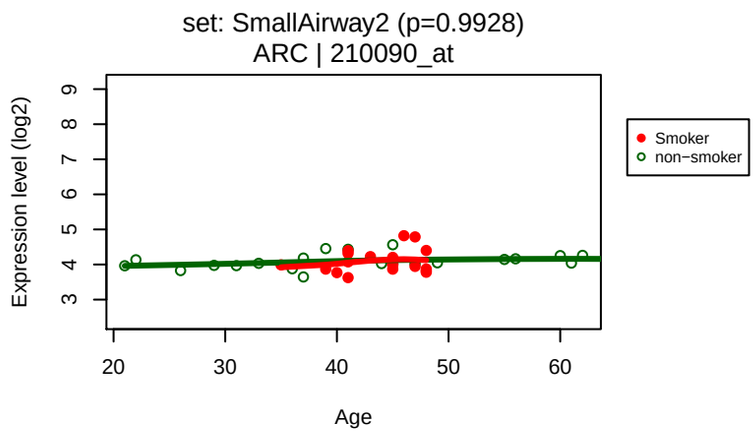
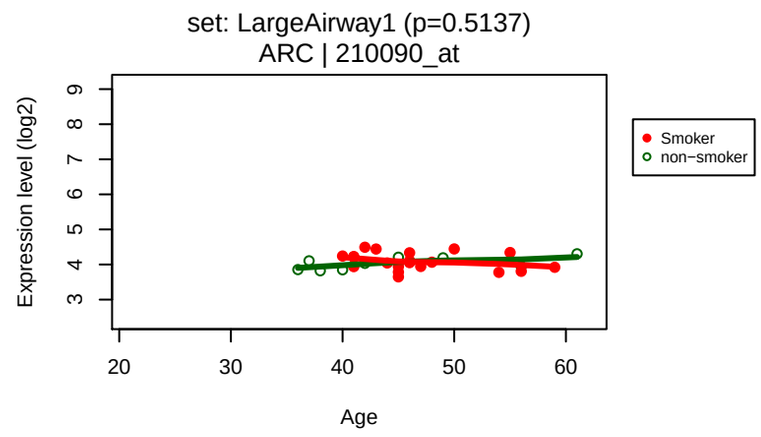

Gene expression of smokers and nonsmokers
gene: ARC | 210090_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1346
p=0.9928
p=0.7603
p=0.5137
p=0.1277
p=0.067
p=0.16
p=0.2928
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

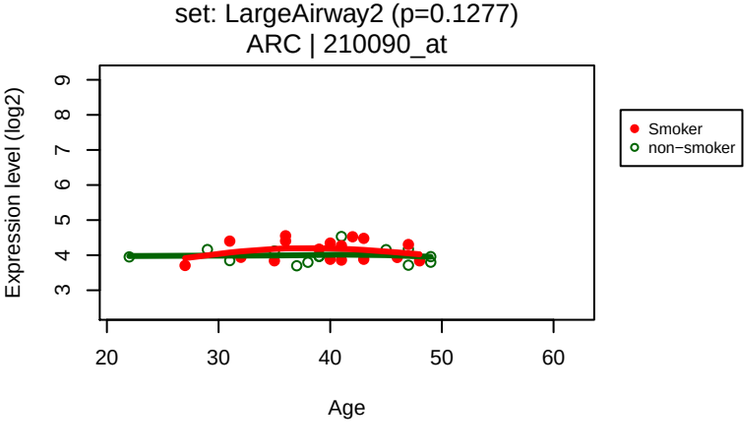

Gene expression ratio (smokers/nonsmokers)
gene: ARC | 210090_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1346
p=0.9928
p=0.7603
p=0.5137
p=0.1277
p=0.067
p=0.16
p=0.2928
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

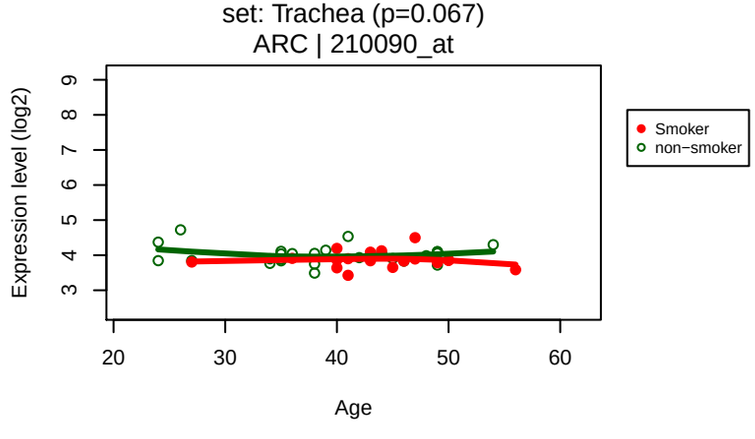

Gene expression ratio in Males
gene: ARC | 210090_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6112
p=0.0125
p=0.1433
p=0.3319
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARC | 210090_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5025
p=0.6584
p=0.5596
p=0.5592
SmallAirway2
LargeAirway2
Trachea
Macrophage2

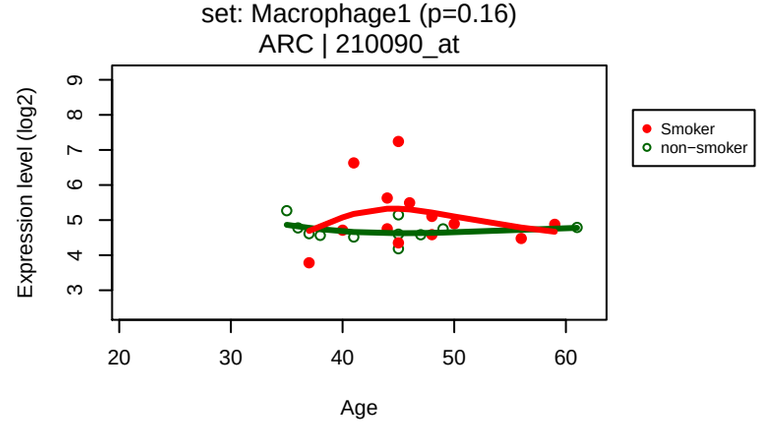

set: SmallAirway1 (p=0.1346)
ARC | 210090_at
Expression level (log2)
Age
Smoker
non−smoker

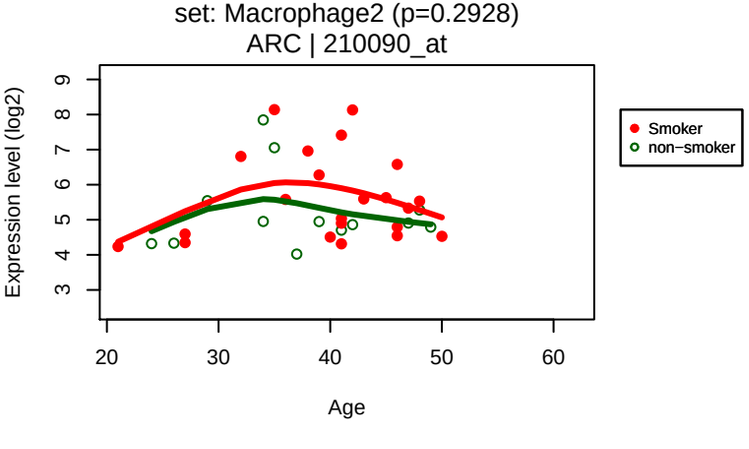

set: SmallAirway2 (p=0.9928)
ARC | 210090_at
Expression level (log2)
Age
Smoker
non−smoker

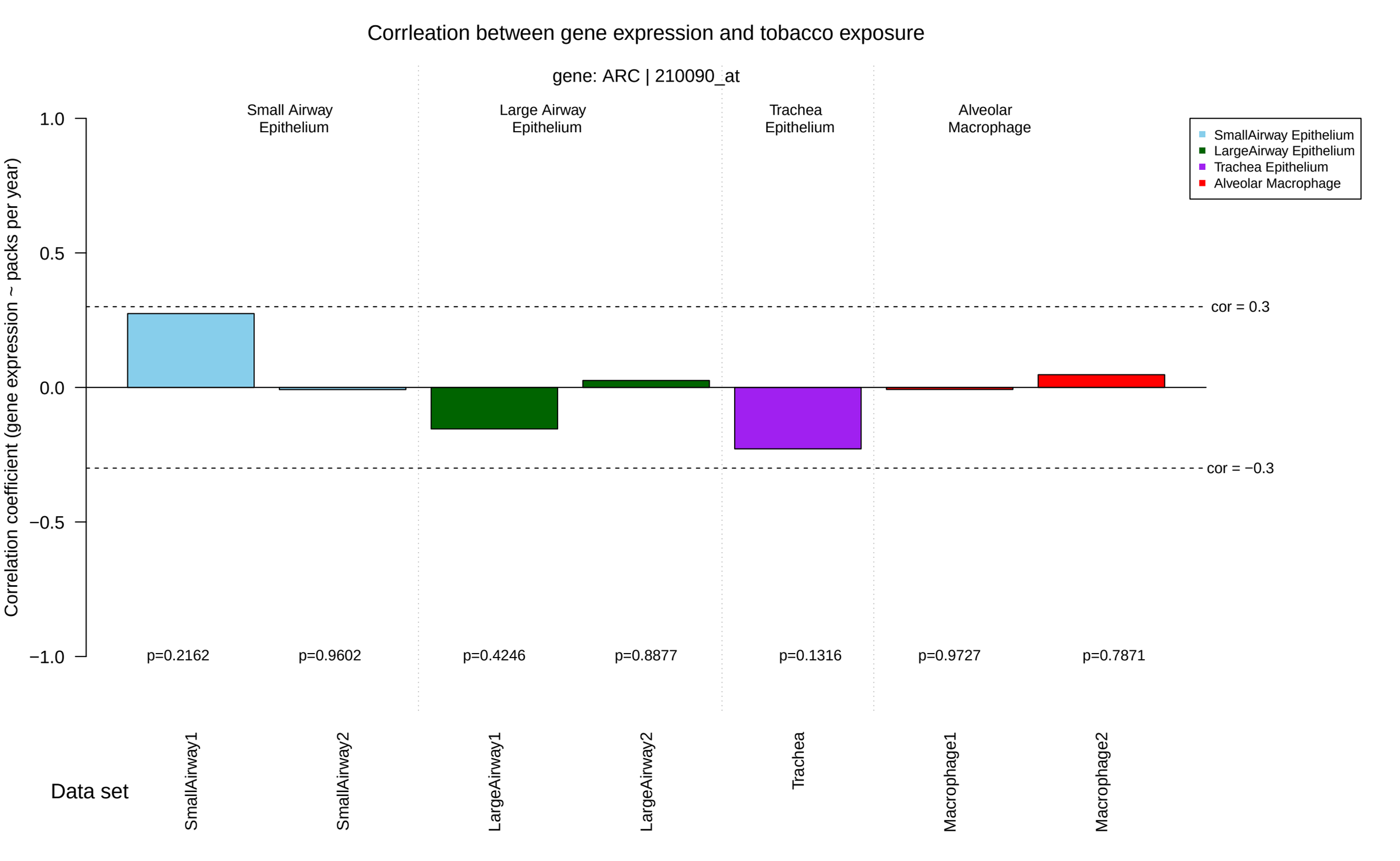

set: LargeAirway1 (p=0.5137)
ARC | 210090_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1277)
ARC | 210090_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.067)
ARC | 210090_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.16)
ARC | 210090_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2928)
ARC | 210090_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARC | 210090_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2162
p=0.9602
p=0.4246
p=0.8877
p=0.1316
p=0.9727
p=0.7871
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2